

gsm8k: accuracy and latency (all budgets)

GSM8K exact match, FullKV 93.10 (cudagraph; deltas vs this row). Same budgets as the narrativeqa slide. 1,319 problems, 100 to 200-token prompts and about 450-token reasoning traces (2,048-token cap), so from 3,072 up no request can cross and every row is FullKV plus run-to-run noise.

Budget	Cross %	vk acc (d)	va acc (d)	vk ttft (s)	va ttft (s)	vk e2e (s)	va e2e (s)
FullKV (eager)	0%	92.87	92.87	35	35	53	53
FullKV (cudagraph)	0%	93.10	93.10	28	28	41	41
1,024	8%	92.6 (-0.5)	93.6 (+0.5)	27 (-2%)	28 (+1%)	41 (-0%)	42 (+2%)
2,048	2%	92.9 (-0.2)	93.1 (+0.0)	27 (-2%)	29 (+4%)	41 (-1%)	43 (+4%)
3,072	0%	92.9 (-0.2)	93.2 (+0.1)	27 (-2%)	29 (+4%)	41 (-1%)	43 (+4%)
4,096	0%	92.5 (-0.6)	91.9 (-1.2)	27 (-1%)	29 (+5%)	42 (+0%)	44 (+5%)
6,144	0%	92.6 (-0.5)	93.3 (+0.2)	27 (-1%)	29 (+5%)	42 (+0%)	43 (+4%)
8,192	0%	93.3 (+0.2)	92.8 (-0.3)	27 (-2%)	29 (+4%)	41 (+0%)	43 (+4%)
11,264	0%	93.0 (-0.1)	92.9 (-0.2)	27 (-2%)	29 (+4%)	41 (-1%)	43 (+4%)
13,664	0%	93.1 (+0.0)	92.9 (-0.2)	27 (-1%)	29 (+4%)	41 (-0%)	43 (+4%)
16,384	0%	93.0 (-0.1)	92.9 (-0.2)	27 (-3%)	29 (+5%)	41 (-1%)	43 (+5%)

Mirror image of narrativeqa: there the 30k-token prompt made every request cross at every budget; here the whole request (prompt plus trace) is under 2,300 tokens, so from 3,072 up cross % is 0 and nothing is evicted; those rows are pure run-to-run noise and calibrate it: 91.9 to 93.3 (batched greedy decoding is not bit-reproducible), so differences under about 1 point anywhere in this deck are noise. ttft/e2e are flat (+4 to 5% for value_attention is the scoring hook's cost, no eviction to pay it back). Only 1,024 and below evict; the budgets that separate the two metrics are on the next slide.

gsm8k: accuracy and latency (all budgets) (256 to 1,024)

GSM8K exact match, FullKV 93.10 (cudagraph; deltas vs this row). 1,319 problems, 100 to 200-token prompts, about 450-token reasoning traces, so cross % falls with the budget. Both metrics prompt protected, evict every 16 steps, 128-token local window, piecewise CUDA graphs.

Budget	Cross %	vk acc (d)	va acc (d)	vk ttft (s)	va ttft (s)	vk e2e (s)	va e2e (s)
FullKV (eager)	0%	92.87	92.87	35	35	53	53
FullKV (cudagraph)	0%	93.10	93.10	28	28	41	41
256	99%	74.8 (-18.3)	77.2 (-15.9)	51 (+86%)	55 (+102%)	82 (+97%)	88 (+113%)
384	71%	87.8 (-5.3)	91.4 (-1.7)	32 (+16%)	30 (+10%)	50 (+20%)	46 (+10%)
512	40%	92.0 (-1.1)	93.0 (-0.1)	28 (+2%)	28 (+1%)	43 (+4%)	42 (+2%)
640	26%	92.3 (-0.8)	93.3 (+0.2)	27 (-0%)	29 (+5%)	42 (+2%)	43 (+5%)
768	17%	92.9 (-0.2)	93.1 (+0.0)	27 (-0%)	29 (+6%)	42 (+1%)	44 (+6%)
1,024	8%	92.6 (-0.5)	93.6 (+0.5)	27 (-2%)	28 (+1%)	41 (-0%)	42 (+2%)

Opposite regime to narrativeqa: prompts are tiny and the trace is the cache, so nothing evicts until the budget drops below the trace length. value_attention is lossless from 640 tokens up (93.3 at 640, 17% of KV reclaimed; 93.0 at 512, 26%) and loses only 1.7 at 384 with 39% reclaimed; vk_ratio needs 768 to be lossless and loses 5.3 at 384. ttft here is queueing (1,319 requests at once), so it tracks the batch's total work; e2e stays within 2 to 6% of FullKV down to 512 and only grows at 384 and 256, where traces lengthen. The load-bearing win is accuracy under eviction.

gsm8k: accuracy and latency (all budgets) (256 to 1,024), evict every 64 steps

Same as the previous slide with SKIVE_EVICT_EVERY=64 SKIVE_EVICT_MARGIN=16: eviction in larger, less frequent rounds. FullKV 93.10, deltas vs cudagraph.

Budget	Cross %	vk acc (d)	va acc (d)	vk ttft (s)	va ttft (s)	vk e2e (s)	va e2e (s)
FullKV (eager)	0%	92.87	92.87	35	35	53	53
FullKV (cudagraph)	0%	93.10	93.10	28	28	41	41
256	96%	79.8 (-13.3)	83.4 (-9.7)	42 (+54%)	42 (+52%)	66 (+59%)	65 (+56%)
384	64%	88.1 (-5.0)	91.9 (-1.2)	32 (+18%)	27 (-0%)	50 (+21%)	42 (+1%)
512	38%	91.3 (-1.8)	92.5 (-0.6)	28 (+2%)	27 (-0%)	43 (+4%)	42 (+0%)
640	24%	91.7 (-1.4)	93.3 (+0.2)	28 (+2%)	27 (-1%)	43 (+4%)	42 (+0%)
768	16%	92.9 (-0.2)	93.3 (+0.2)	27 (-1%)	27 (-1%)	42 (+1%)	41 (-0%)
1,024	8%	92.7 (-0.4)	92.9 (-0.2)	27 (-2%)	27 (-1%)	41 (-0%)	42 (+0%)

Slow cadence is the lever for budgets below the trace length: value_attention 91.9 at 384 (was 91.4) and 83.4 at 256 (was 77.2), vk_ratio 79.8 at 256 (was 74.8), with the same KV reclaimed. It also removes the latency overhead: value_attention e2e is within 1% of FullKV at every budget from 384 up (42 s vs 41 s), because requests run up to 16 blocks over budget between rounds and stop re-deriving the working context.